Ethan Moon

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Education

University of Michigan College of Engineering *Ann Arbor, MI*

Bachelor of Science in Engineering, Major in Computer Science, Minor in Mathematics Expected April 2030

- **Relevant Coursework:** Programming, Discrete Mathematics, Proof-Based Linear Algebra, Multivariable Calculus

Technical Skills

Languages: TypeScript, JavaScript, Python, C++, Swift, SQL
Frameworks & Platforms: React, React Native/Expo, Next.js, Node.js, tRPC, SwiftUI, ROS 2, Isaac Sim, OpenFOAM
Data, Testing & DevOps: PostgreSQL, Supabase, Prisma, Drizzle ORM, Vitest, Playwright, GitHub Actions, Git
Spoken Languages: English & Korean (Native/Bilingual); Spanish (Limited Working)

Experience

Lead Engineer & Head of U.S. Team, Ody *May 2026 – Present*

Early-Stage Restaurant-Tech Startup · Reservations, Ordering & Loyalty Platform Remote · U.S., France & China

- **Product:** Shipped **414 merged pull requests** (4th of 27 engineers), including AI menu translation in **7 languages**, automatic discounts, email marketing, analytics for restaurant chains, and a customer app redesign.
- **Leadership:** Promoted from intern to lead in **3 months** (ranked first of six); lead a **10+ person U.S. team** of Stanford, Oxford, CMU, and Berkeley students alongside teams in France and China, and run **10+ technical interviews**.
- **AI:** Lead developer of an AI tool that builds restaurant websites in 8 languages with custom domains and a chat assistant. Wrote **74%** of its code and raised Google search (SEO) scores from **81 to 96 out of 100**.
- **Speed:** Made restaurant pages load over **2× faster** (4.6 s to 2.1 s), cut the online-ordering widget's code by **57%**, sped up SEO checks by **77%**, and shrank the iOS app by **16.5%**.

Software Engineer, REDDY *September 2026 – Present*

Career-Fair Recruiting App · Michigan Build & Launch Ann Arbor, MI

- Wrote the product spec for an event-success score after researching **8 products**, including Microsoft Clarity, then built it into the organizer dashboard and printable manager reports, combining **6 database queries into 1**.
- Cut the student events page's server requests by **94%** (125 to 8) and its data by **96%** (1.4 MB to 58 KB), and replaced constant refreshing with live updates, removing ~**240 requests per hour** per recruiter.

Software Engineer, VOID Tech *October 2026 – Present*

Student Technology Consulting Ann Arbor, MI

- Top contributor on a **5-person** team building an online store and staff portal for a New York specialty grocer, writing about **60%** of the code ahead of launch.
- Turned the owners' requirements into **9 settings sections** (shipping, pickup-only perishables, New York sales tax) and built order management and a secure staff login.

Undergraduate Researcher, University of Michigan *August 2026 – Present*

Intelligent Robotics & Autonomy Laboratory (iRaL) · Prof. Vasileios Tzoumas Ann Arbor, MI

- Built the simulation and control software (Isaac Sim, ROS 2) for a six-joint robotic arm in the lab's morphable-drone research, plus a web dashboard to operate it, writing **5,000+ lines** and **54%** of the lab's code.
- Programmed motion control that updates **60 times per second** with as little as **1.9 mm** error, added demo recording, replay, and emergency-stop safety, and designed the interface the lab adopted to link simulation to the real robot.

Researcher & Co-Author, Grand Valley State University *September 2025 – September 2026*

- Ran **20** computer simulations (OpenFOAM) of thick bio-inks and food inks flowing through a 3D-printer nozzle, using **56 processor-hours**, and confirmed results stayed stable at finer detail (under **0.05%** change).
- Built a dashboard that lets the research team explore simulation results and trace where each number came from, and led a review that caught and fixed two physics errors in the analysis.
- Authored **3 papers**: NAMRC 55 (**first author**, submitted), ASEE 2026 (published), and a bioink paper (in prep.).

Projects

NoSpoilers – Spoiler-Free Movie Discovery Website & Chrome Extension *June 2026 – Present*

- Co-built and launched a spoiler-free movie discovery site with **150+ users**, writing **58 of its 61** merged pull requests and **82%** of the code added since the first prototype.
- Single-handedly built a Chrome extension that hides spoilers across the web, plus an AI recap assistant that answers questions using only the scenes a viewer has already watched.

Honors & Awards

DECA State Champion · Dartmouth Book Award · **2nd Place, Engineering**, GYM International STEM Innovation Challenge · Finalist, International Astronomy & Astrophysics Competition